Cluster Before You Train: Fingerprint-Based Concept Discovery for Federated Learning Under Recurrent Drift

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Abstract

We introduce a *protocol ordering* perspective for clustered federated learning under concept drift: the choice of *when* in the federation round to cluster clients—before or after local training—determines clustering quality in the concept-recurrence regime. Existing methods cluster in weight space *after* training, but under recurrence, stale model initializations contaminate the clustering signal, producing an empirically observed clustering-error floor $\eta \geq 0.30$ that no post-training algorithm in our benchmark breaches when concept separation is meaningful (Oracle gap $>$ 4pp). We formalise this in a one-step linearised Gaussian model: stale model initialisations inflate Scheme A clustering error to first order in concept-switching variance, and a d/L_{cls} SNR calculation suggests—consistent with our empirical observation of a $200\times$ gap—that data fingerprints achieve lower error. We instantiate this in **FedPro-Track** (FPT)—a three-phase protocol combining class-conditional fingerprints, OT spectral clustering, and empirical-Bayes shrinkage. Across 4 datasets (including natural temporal drift), 26 baselines from 7 method families, and 60+ experimental runs, FPT achieves accuracy within 0.3pp of Oracle on CIFAR-100 and $+6.9$pp above the best existing baseline on fMoW, while all cluster-after-train methods rank below FedAvg on every dataset where the Oracle gap exceeds 4pp.

1 Introduction

Federated learning (FL) enables collaborative model training without sharing raw data [McMahan et al., 2017, Kairouz et al., 2021]. In many deployments, client data distributions shift over time due to *concept drift* [Gama et al., 2014, Lu et al., 2018]: each client experiences one of C latent concepts that change across rounds [Jothimurugesan et al., 2023, Panchal et al., 2023]. When the concept structure is known, the server can maintain separate models per group, eliminating cross-concept interference. This premise has motivated a rich line of clustered FL methods [Ghosh et al., 2020, Sattler et al., 2021, Marfoq et al., 2021, Guo et al., 2024, Licciardi et al., 2025]. Yet deploying these methods in the *concept-recurrence* regime—where clients revisit previously-seen concepts—raises three open challenges.

Challenge 1: When to cluster. Existing theory characterises *whether* concept-level aggregation outperforms global averaging [Chen et al., 2023, Yu et al., 2025, Mansour et al., 2020], but not *when in the federation round* to make the clustering decision. The dominant approach clusters *after* local training, using weight updates Δ_k as the signal [Long et al., 2023, Kim et al., 2024]. A few methods use data summaries before training [Yang et al., 2023], but lack theoretical analysis of why pre-training clustering should be preferred and do not address stale initialisation. Is post-training clustering the right protocol ordering?

Challenge 2: Stale-initialization contamination. Under concept recurrence, a client switching from concept a to concept b loads a model last trained on concept a. Its weight update Δ_k conflates the true concept-b gradient with a stale-initialization mismatch unrelated to concept identity. We observe that this contamination produces a clustering-error lower bound $\eta \geq 0.30$ across all cluster-after-train methods on every dataset where the Oracle gap exceeds 4pp (Table 1).

Challenge 3: Adaptive aggregation under unknown SNR. Even with correct clustering, the benefit of concept-level aggregation depends on the per-concept signal-to-noise ratio (SNR), which is rarely known

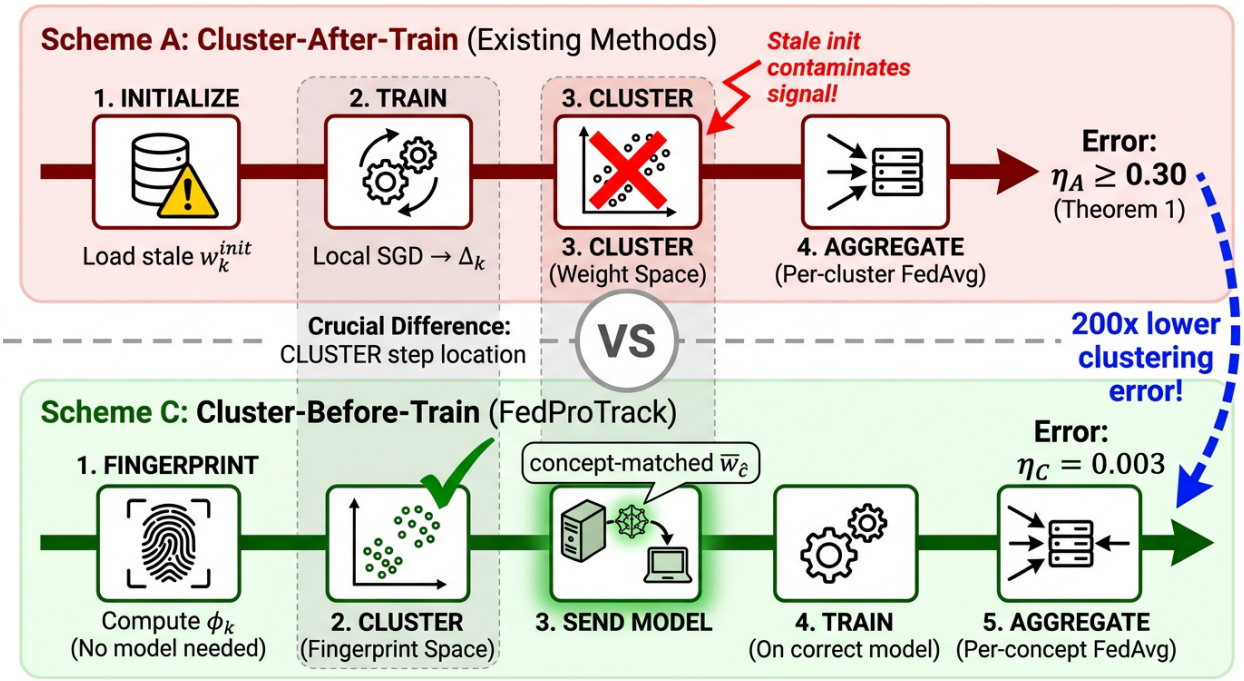

Figure 1: **Protocol ordering: the core design choice.** *Top* (Scheme A, red): existing methods cluster clients *after* local training—stale model initialisations contaminate weight updates, yielding an empirically observed clustering-error floor $\eta_A \geq 0.30$ across the datasets we study (cf. Theorem 4). *Bottom* (Scheme C, green): FedProTrack clusters *before* training using lightweight data fingerprints, then sends concept-matched models to clients, achieving $\eta_C = 0.003$—a $200\times$ improvement.

in advance. A method that commits fully to concept-level models when SNR is low can *increase* variance beyond the bias it eliminates.

We address these challenges in **FedProTrack** (FPT), a modular protocol that clusters *before* training using data fingerprints (Figure 1). FPT computes class-conditional data statistics (C1), applies OT spectral clustering with automatic concept-count selection (C1), loads concept-matched models from a memory bank (C2), and interpolates toward the global mean via empirical-Bayes shrinkage (C3)—all at $2.1\times$ FedAvg communication. Across 4 datasets and 26 baselines, FPT achieves accuracy within 0.3–1.5pp of Oracle on all datasets where the Oracle gap exceeds 4pp (Table 1).

Contributions.

1. **Stale-initialisation amplification theorem** (§4): In a one-step linearised Gaussian regime, we prove that stale model initialisations inflate Scheme A clustering error by $\geq \kappa \cdot \mathbb{V}(w^{(\text{init})})$, and show via a d/L_{cls} SNR analysis (Lemma 6) that fingerprint clustering achieves lower error—verified empirically as a $200\times$ gap. Combined with an implicit-shrinkage characterisation (Corollary 2), this yields an excess-risk ordering.

2. **FedProTrack** (§4): A Scheme C protocol with fingerprint-based concept discovery, concept-matched model loading, and adaptive shrinkage that addresses all three challenges within a unified framework.

3. **Comprehensive validation** (§5): 4 datasets (CIFAR-100, Fashion-MNIST, fMoW with natural drift, CIFAR-10 as negative control), 26 baselines, 7 method families. FPT matches Oracle while every cluster-after-train baseline exhibits $\eta \geq 0.30$ on datasets where concept-level treatment is beneficial (Oracle gap

> 4pp).

2 Related Work

Clustered FL. Most methods cluster clients *after* local training using weight-space signals: IFCA [Ghosh et al., 2020], CFL [Sattler et al., 2021], FedEM [Marfoq et al., 2021], FedRC [Guo et al., 2024], FedGWC [Licciardi et al., 2025], and gradient-based partitioning [Kim et al., 2024]. A few methods cluster before training using data summaries—e.g., FedTCM [Yang et al., 2023] uses label-count vectors—but these lack theoretical analysis of why pre-training clustering is preferable, do not address stale-initialisation contamination, and are limited to label-histogram similarity. Our work provides the first theoretical justification (Theorem 4) and introduces class-conditional fingerprints that capture richer distributional structure.

Personalized FL. Methods such as APFL [Deng et al., 2020], Ditto [Li et al., 2021b], pFedMe [T Dinh et al., 2020], FedBN [Li et al., 2021c], and FedPer [Arivazhagan et al., 2019] mix local and global models. FedRep [Collins et al., 2021], FedBABU [Oh et al., 2022], MOON [Li et al., 2021a], FedProto [Tan et al., 2022], and FedALA [Zhang et al., 2023] share representations or prototypes. None maintain concept-level models or exploit concept recurrence through memory reuse.

FL under concept drift. FedDrift [Jothimurugesan et al., 2023] and Flash [Panchal et al., 2023] detect and adapt to drift; FedCCFA [Chen and Chao, 2024] addresses concept shift in online FL. None characterise *when* concept-level treatment is worth the variance cost. Concept drift has been extensively studied in centralised settings [Gama et al., 2014, Lu et al., 2018]; our work brings the when-to-cluster question to the federated setting.

Federated optimization. FedProx [Li et al., 2020] adds a proximal term; SCAFFOLD [Karimireddy et al., 2020] uses control variates; adaptive methods [Reddi et al., 2021, Wu et al., 2023] apply server-side momentum; FedSR [Nguyen et al., 2022] targets domain generalisation. These address statistical heterogeneity but do not model discrete concept recurrence.

Theory. Chen et al. [2023] establish minimax bounds for the global-vs-local threshold; Yu et al. [2025] extend these to communication-accuracy tradeoffs. The implicit-shrinkage connection (Corollary 2) relates to James-Stein shrinkage [James and Stein, 1961, Efron and Morris, 1973, Sontakke et al., 2024] but has not been formalised for clustered FL. Spectral clustering consistency [Lei and Rinaldo, 2015, Löffler et al., 2021, Von Luxburg, 2007] underpins our Phase A guarantees.

3 Preliminaries

3.1 Problem Setting

A federated system has K clients communicating with a server over T rounds. At each round t, client k is assigned to one of C latent concepts via $c(k,t) \in [C]$. Client k observes n i.i.d. samples (x_i, y_i) from a concept-specific distribution $P_{c(k,t)}$. Concepts may *recur*: a client assigned to concept a at round t may return to concept a at a later round $t' > t$ after visiting other concepts.

We analyse this setting under a canonical linear model: $y = \langle w_j^*, x\rangle + \varepsilon$, where $w_j^* \in \mathbb{R}^d$ is the concept-j parameter, $x \sim \mathcal{N}(0, \Sigma)$, and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The bias of concept j relative to the global mean is $B_j^2 := \|w_j^* - \bar{w}^*\|_\Sigma^2$, where $\bar{w}^* = \frac{1}{C}\sum_j w_j^*$.

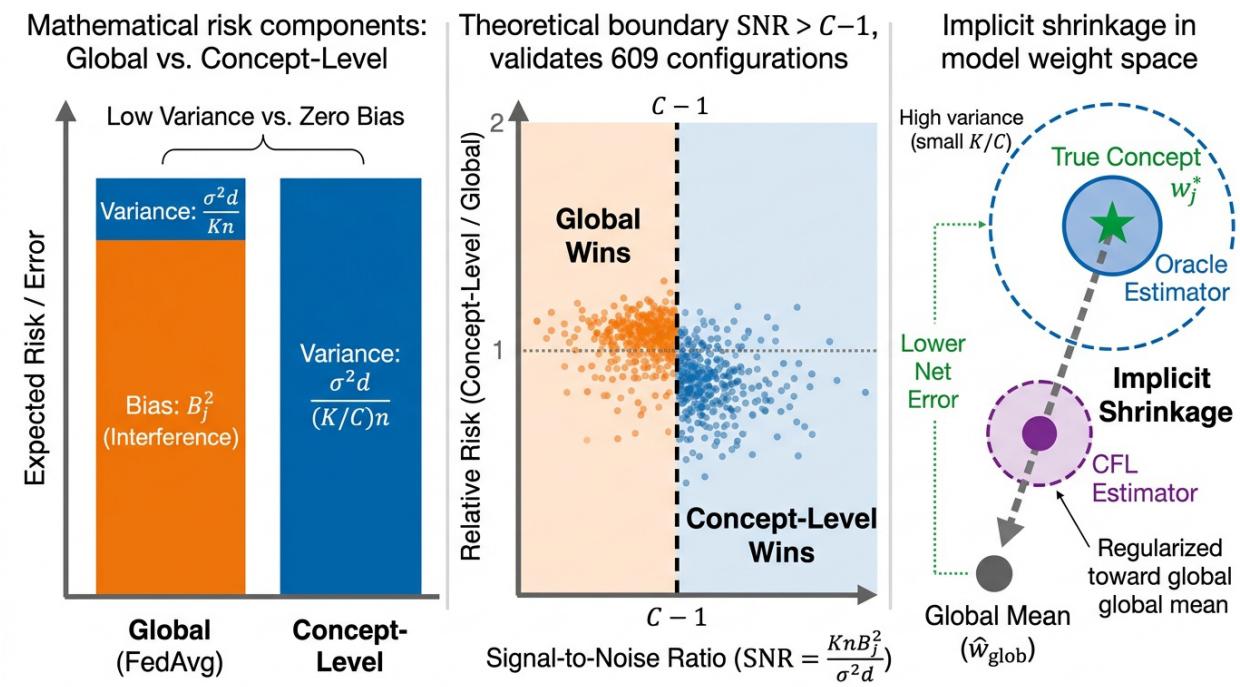

Figure 2: The crossover condition. *Left*: bias-variance tradeoff. *Center*: sharp threshold $\text{SNR} > C-1$, validated on 609 synthetic configurations. *Right*: imperfect clustering as implicit shrinkage—moderate η can outperform Oracle when K/C is small.

3.2 When Does Concept-Level Aggregation Help?

Proposition 1 (Bias-Variance Decomposition)**.** *Under balanced concepts ($|\mathcal{C}_j| = K/C$) and isotropic design ($\Sigma = I_d$):*

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_{\text{glob}})] = B_j^2 + \frac{\sigma^2 d}{Kn}, \qquad \mathbb{E}[\mathcal{E}_j(\hat{w}_j)] = \frac{\sigma^2 C d}{Kn}. \tag{1}$$

Concept-level aggregation outperforms global aggregation iff $\text{SNR}_{\text{concept}} := KnB_j^2/(\sigma^2 d) > C-1$. This crossover condition extends to unbalanced proportions, noisy labels, and anisotropic features (Appendix D).

Implicit shrinkage from imperfect clustering. Any practical clustering algorithm introduces label noise.

Corollary 2 (Implicit Shrinkage)**.** *Under balanced concepts and the Gaussian linear model, a clusterer with symmetric error rate η produces an estimator whose bias is equivalent to shrinkage with coefficient $\lambda_{\text{imp}} = \eta C/(C-1)$ toward the global mean $\bar{w}^*$ [Efron and Morris, 1973, James and Stein, 1961]. This is a first-order characterisation; the exact finite-sample relationship depends on cluster sizes and loss geometry.*

This explains why CFL [Sattler et al., 2021] can match Oracle when K/C is small: the regularisation from imperfect clustering compensates for bias. As K grows, Oracle's advantage dominates (Appendix I). The key implication is that *reducing η is the primary lever* for concept-level FL—motivating the protocol ordering analysis in the next section.

4 Method: FedProTrack

We first establish *why* clustering before training is advantageous—through a formal analysis in a one-step linearised Gaussian model and corroborating empirical evidence—in §4.1, then describe FPT's four modules (Figure 3): fingerprint extraction (§4.2), concept discovery (§4.3), concept-matched training (§4.4), and adaptive shrinkage (§4.5).

4.1 Theoretical Foundation: Protocol Ordering

Existing clustered FL methods share a common protocol structure that we term **Scheme A** (cluster-after-train):

Definition 3 (Protocols)**.** *Fix a round t. Each protocol takes concept models $\{\bar{w}_c^{(t-1)}\}_{c=1}^{C}$ and outputs updated models.*

Scheme A *(cluster-after-train): (i) each client loads a default initialization $w_k^{(\text{init})}$ and runs local SGD to produce $\Delta_k^{(t)}$; (ii) the server clusters $\{\Delta_k^{(t)}\}$ and assigns labels $\hat{c}_k^A$; (iii) per-cluster averaging.*

Scheme C *(cluster-before-train): (i) each client computes and sends a data fingerprint $\phi_k^{(t)}$ (no training needed); (ii) the server clusters $\{\phi_k^{(t)}\}$ and assigns labels $\hat{c}_k^C$; (iii) each client receives $\bar{w}_{\hat{c}_k^C}^{(t-1)}$ and runs local SGD; (iv) per-cluster averaging.*

The critical difference is *what each client loads before training*. In Scheme A, the initialization $w_k^{(\text{init})}$ is typically the last model the client used—which, under concept recurrence, was trained on a *different* concept. In Scheme C, the server first identifies the client's current concept from its fingerprint and sends the *concept-matched* model, eliminating the stale-initialization mismatch entirely.

Assumptions. We state the regularity conditions: (A1) L-smooth, μ-strongly-convex per-concept losses; (A2) bounded gradient noise $\mathbb{E}\|\nabla L_j(w) - g\|^2 \leq \sigma_g^2$; (A3) concept separation $\|w_a^* - w_b^*\|_\Sigma \geq \delta > 0$ for $a \neq b$; (A4) norm-based clustering metric on weight updates; (A4′) fingerprint separation $\|\phi(P_a) - \phi(P_b)\| \geq \beta > 0$; (A5) fingerprint-space clustering is at least as accurate as clean weight-space clustering, $\eta_C \leq \eta_A^{(\text{clean})}$, motivated by the d/L_{cls} SNR analysis of Lemma 6 (where L_{cls} denotes the number of classes) and verified empirically on CIFAR-100 (Appendix C).

Theorem 4 (Stale-Initialisation Amplification)**.** *Let η_A be the clustering error of Scheme A and $\eta_A^{(\text{clean})}$ the error of Scheme A with Oracle-provided correct initialisation (i.e., $\mathbb{V}(w^{(\text{init})}) = 0$). Assume (A1–A4), a single local SGD step under the Gaussian linear model of Lemma 6 (frozen backbone, linear head), and that the stale-initialisation perturbation appears in the clustering input as approximately isotropic additive variance $\mathbb{V}(w^{(\text{init})})$ (a pairwise / one-vs-rest reduction to the two-class setting of Lemma 5, with inter-concept separation given by (A3)). Then, to first order in $\mathbb{V}(w^{(\text{init})})$:*

$$\eta_A \;\geq\; \eta_A^{(\text{clean})} \;+\; \kappa \cdot \mathbb{V}(w^{(\text{init})}), \tag{2}$$

where $\mathbb{V}(w^{(\text{init})}) := \mathbb{E}[\|w_k^{(\text{init})} - \bar{w}_{c(k,t)}^{(t-1)}\|^2 \mid c(k,t)]$ is the stale-initialisation variance and $\kappa > 0$ depends on the Lipschitz constant L, the concept separation δ, and the per-class covariance of Lemma 5. If additionally (A5) holds—i.e., $\eta_C \leq \eta_A^{(\text{clean})}$—then

$$\eta_A \;\geq\; \eta_C \;+\; \kappa \cdot \mathbb{V}(w^{(\text{init})}). \tag{3}$$

Inequality (2) is a first-order statement within the one-step linearised regime: stale initialisation inflates Scheme A clustering error by $\geq \kappa \cdot \mathbb{V}$, with higher-order corrections in $\mathbb{V}$. Inequality (3) additionally

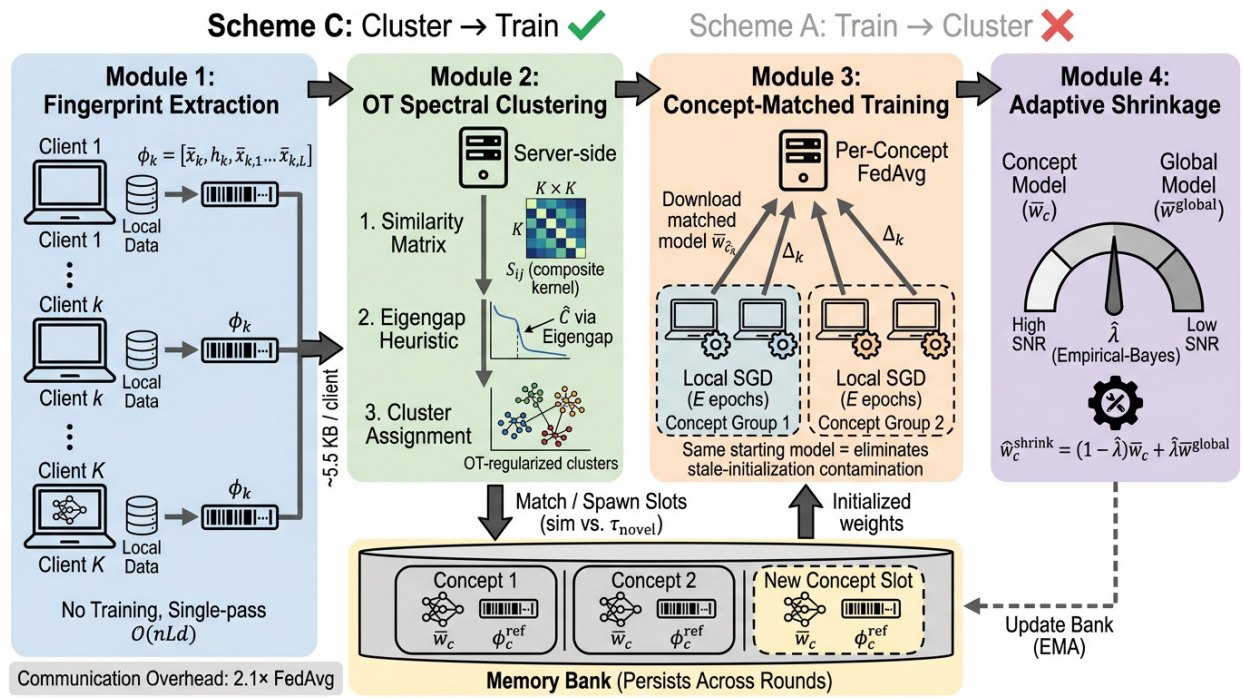

Figure 3: **FedProTrack pipeline (one federation round).** Module 1 (blue): each client computes a class-conditional fingerprint ϕ_k in a single pass ($\mathcal{O}(dL)$ bytes, no training). Module 2 (green): the server performs OT spectral clustering with eigengap-based concept-count selection and matches clusters to a persistent memory bank. Module 3 (orange): clients receive concept-matched models and perform local training; per-concept FedAvg aggregation follows. Module 4 (purple): empirical-Bayes shrinkage interpolates between concept-level and global models based on estimated SNR. Communication overhead: $2.1\times$ FedAvg.

invokes (A5); the d/L_{cls} SNR calculation of Lemma 6 gives theoretical support for (A5), which we verify empirically ($\eta_A^{(\text{clean})} = 0.40$ vs. $\eta_C = 0.002$ on CIFAR-100; Appendix C).

Proof sketch. By L-smoothness (A1) and Taylor expansion, Δ_k decomposes into a concept signal, a Hessian-scaled stale-init bias $\alpha H_k(w_k^{(\text{init})} - \bar{w}_{c(k,t)}^{(t-1)})$ with $\|H_k\| \leq L$, and higher-order terms. The bias adds variance $\mathbb{V}$ to the norm-based clustering input (A4); by Lemma 5, the Bayes-optimal error is monotonically increasing in input variance, yielding (2). Full proof: Appendix A. $\mathbb{V} > 0$ whenever concepts recur and clients do not receive concept-matched models; when $\mathbb{V} = 0$, the gap vanishes (Appendix G).

From clustering error to excess risk. Under the balanced Gaussian model, combining (3) with Corollary 2 yields a first-order heuristic under which the excess-risk gap scales as $(\eta_A - \eta_C) \cdot \frac{C}{C-1} \cdot \bar{B}^2$, where $\bar{B}^2 := \frac{1}{C}\sum_{j=1}^{C} \|w_j^* - \bar{w}^*\|_\Sigma^2$ is the mean between-concept bias. A tight finite-sample lower bound is left to future work.

4.2 Fingerprint Extraction

Each client k computes a *fingerprint* $\phi_k \in \mathbb{R}^p$ from its local data *without any model training*. The fingerprint concatenates three statistics that jointly capture the data distribution:

$$\phi_k = \Big[\ \underbrace{\bar{x}_k}_{\text{global mean}}\ ,\ \underbrace{h_k}_{\text{label dist.}}\ ,\ \underbrace{\bar{x}_{k,1}, \ldots, \bar{x}_{k,L_{\text{cls}}}}_{\text{class-conditional means}}\ \Big], \tag{4}$$

where $\bar{x}_k = \frac{1}{n}\sum_i x_i^{(k)} \in \mathbb{R}^d$ is the global feature mean computed over all n local samples, $h_k \in \mathbb{R}^{L_{\text{cls}}}$ is the normalised label histogram (L_{cls} is the number of classes; notation distinct from the Lipschitz constant L of (A1)), and $\bar{x}_{k,\ell} = \frac{1}{n_\ell}\sum_{i:\, y_i=\ell} x_i^{(k)} \in \mathbb{R}^d$ is the mean feature vector for class ℓ (zero-padded when class ℓ is absent). For ResNet-18 [He et al., 2016] frozen features with $d = 128$ and $L_{\text{cls}} = 10$ classes, $p = d + L_{\text{cls}} + L_{\text{cls}} \cdot d = 1418$ floats (≈5.5 KB per client).

The fingerprint design is motivated by two properties. First, class-conditional means are *sufficient statistics* for the concept identity under Gaussian class distributions—two clients sharing the same concept will produce similar $\bar{x}_{k,\ell}$ vectors, while clients with different class subsets or feature distributions will diverge. Second, the fingerprint is computed in a single pass over local data with $O(nLd)$ cost and requires no gradient computation or model access, making it communication-efficient (only aggregate statistics leave the client, though a formal privacy analysis is beyond the scope of this work).

4.3 Concept Discovery via OT Spectral Clustering

Given fingerprints $\{\phi_1, \ldots, \phi_K\}$, the server discovers concepts in three steps.

Step 1: Similarity computation. The server constructs a pairwise similarity matrix $S \in \mathbb{R}^{K\times K}$ using a composite kernel that combines feature-level, label-level, and class-conditional similarities:

$$S_{ij} = \alpha_1 \,\text{sim}_{\text{feat}}(\phi_i, \phi_j) + \alpha_2 \,\text{sim}_{\text{label}}(h_i, h_j) + \alpha_3 \,\text{sim}_{\text{cc}}(\phi_i, \phi_j), \tag{5}$$

with fixed weights $(\alpha_1, \alpha_2, \alpha_3) = (0.25, 0.30, 0.45)$, set on a held-out synthetic dataset before benchmark evaluation and not tuned on any evaluation dataset. Here sim_{feat} applies a Gaussian kernel to the global-mean distance, $\text{sim}_{\text{label}}$ computes cosine similarity of label histograms, and sim_{cc} averages Gaussian-kernel similarities of class-conditional means over shared classes. The dominant weight on sim_{cc} reflects that class-conditional structure carries the most discriminative signal for concept identity. FPT is robust to these weights: accuracy varies by ≤ 0.2pp and η_C is unchanged across four non-degenerate configurations spanning uniform, inverted, and CC-dominant weightings (3 seeds each; Appendix K).

Step 2: Automatic concept-count selection. The number of concepts $\hat{C}$ is determined by an eigengap heuristic on the normalised Laplacian $L_{\text{sym}} = I - D^{-1/2}SD^{-1/2}$ [Von Luxburg, 2007]: we select $\hat{C} = \arg\max_{c\in[2,K/2]}(\lambda_{c+1} - \lambda_c)$, where $\lambda_1 \leq \ldots \leq \lambda_K$ are the sorted eigenvalues. This eliminates the need to specify C as a hyperparameter.

Step 3: OT-regularised spectral clustering. Spectral clustering on the top-$\hat{C}$ eigenvectors of L_{sym} partitions clients into groups $\{\mathcal{C}_1, \ldots, \mathcal{C}_{\hat{C}}\}$. To avoid degenerate cluster sizes, we replace the standard k-means step with an optimal-transport (OT) assignment [Villani, 2009] that enforces a minimum cluster size of $\lfloor K/(2\hat{C})\rfloor$.

Similarity calibration and memory matching. The server maintains a *memory bank* $\mathcal{M}$ that stores a model checkpoint $\bar{w}_c$ and reference fingerprint ϕ_c^{ref} per concept. After clustering, each group is matched to the closest memory slot by fingerprint similarity. To handle novel concepts, FPT computes a novelty threshold τ_{novel} as the $(1-\alpha)$-quantile ($\alpha = 0.05$) of the pairwise fingerprint similarity distribution. When the best-matching memory slot has similarity below τ_{novel}, a new concept slot is spawned with a freshly initialised model.

4.4 Concept-Matched Training and Aggregation

This module implements the key operational insight of Scheme C: each client trains on a model that *matches its current concept*, eliminating the stale-initialization variance $\mathbb{V}(w^{(\text{init})})$.

For each client k assigned to concept $\hat{c}_k$, the server sends the concept-matched model $\bar{w}_{\hat{c}_k}^{(t-1)}$ from the memory bank. The client runs E epochs of local SGD on its data $(x_i, y_i) \sim P_{c(k,t)}$ starting from $\bar{w}_{\hat{c}_k}^{(t-1)}$,

producing an update $\Delta_k^{(t)} = w_k^{(t)} - \bar{w}_{\hat{c}_k}^{(t-1)}$. The server then performs per-concept federated averaging:

$$\bar{w}_c^{(t)} = \bar{w}_c^{(t-1)} + \frac{1}{|\mathcal{C}_c|} \sum_{k \in \mathcal{C}_c} \Delta_k^{(t)}. \tag{6}$$

Because all clients in $\mathcal{C}_c$ start from the *same* concept model $\bar{w}_c^{(t-1)}$, their updates $\Delta_k^{(t)}$ reflect only the concept-c gradient signal—free of the cross-concept contamination that plagues Scheme A. This is the mechanism by which Theorem 4's theoretical advantage becomes operational.

4.5 Adaptive Shrinkage

Even with correct concept assignments, the benefit of concept-level aggregation depends on whether the per-concept SNR exceeds the crossover threshold $C-1$ (Proposition 1). When the SNR is low—e.g., when concepts differ only mildly or when per-concept sample sizes $|\mathcal{C}_c| \cdot n$ are small—concept-level estimators suffer from high variance.

FPT addresses this with an *empirical-Bayes shrinkage* step that interpolates between concept-level and global models:

$$\hat{w}_c^{\text{shrink}} = (1 - \hat{\lambda})\, \bar{w}_c^{(t)} + \hat{\lambda}\, \bar{w}^{(t)}, \tag{7}$$

where $\bar{w}^{(t)} = \frac{1}{\hat{C}} \sum_c \bar{w}_c^{(t)}$ is the global mean model and the shrinkage coefficient is estimated as

$$\hat{\lambda} = \frac{\hat{\sigma}^2/((K/\hat{C})n)}{\hat{\sigma}^2/((K/\hat{C})n) + \hat{\sigma}_B^2}, \tag{8}$$

with $\hat{\sigma}^2$ the estimated within-concept noise and $\hat{\sigma}_B^2 = \frac{1}{\hat{C}} \sum_c \|\bar{w}_c^{(t)} - \bar{w}^{(t)}\|^2$ the empirical between-concept variance. When the between-concept signal is strong ($\hat{\sigma}_B^2 \gg \hat{\sigma}^2/((K/\hat{C})n)$), $\hat{\lambda} \to 0$ and the estimator is fully concept-specific. When the signal is weak ($\hat{\sigma}_B^2 \to 0$), $\hat{\lambda} \to 1$ and FPT gracefully degrades to global averaging (FedAvg).

This mechanism connects to the implicit-shrinkage analysis (Corollary 2): while imperfect clustering *unintentionally* regularises via noisy assignments, FPT's shrinkage module provides *intentional*, SNR-adaptive regularisation.

Communication cost. Each round, FPT transmits one fingerprint ($p = 1418$ floats ≈ 5.5 KB) and one model ($d \times L + L = 1290$ params ≈ 5.0 KB) in each direction per client. The total per-round cost is $2.1\times$ FedAvg ($1\times$ model upload $+$ $1\times$ model download $+$ $0.1\times$ fingerprint overhead).

5 Experiments

We design our evaluation to test the three challenges: (1) Does clustering before training close the Oracle gap? (2) Do all cluster-after-train baselines exhibit high η? (3) Does adaptive shrinkage handle low-SNR regimes?

5.1 Setup

Datasets. We evaluate on four benchmarks (K=20, T=100, ρ=25, C=4): CIFAR-100 [Krizhevsky, 2009] (flagship; also with $K \in \{20, 40\}$, $\rho \in \{17, 25, 33\}$, 5 configs), Fashion-MNIST [Xiao et al., 2017], fMoW [Koh et al., 2021] (robustness test with natural temporal drift; C=4 concepts are discretised from continuous temporal evolution spanning 2002–2017, providing a stress test where inter-concept boundaries

Algorithm 1 FedProTrack: One Federation Round

Require: Clients $\{1, \ldots, K\}$, memory bank $\mathcal{M} = \{(\bar{w}_c, \phi_c^{\text{ref}})\}_{c=1}^{C_{\text{prev}}}$

Fingerprint Extraction (client-side, no training):
for each client k **in parallel do**
  $\phi_k \leftarrow [\bar{x}_k,\ h_k,\ \bar{x}_{k,1}, \ldots, \bar{x}_{k,L}]$ {class-conditional fingerprint}
  Send ϕ_k to server
end for
Concept Discovery (server-side):
Build similarity matrix S from $\{\phi_k\}$
Eigengap $\rightarrow \hat{C}$; spectral clustering + OT $\rightarrow$ groups $\{\mathcal{C}_1, \ldots, \mathcal{C}_{\hat{C}}\}$
Match groups to memory bank $\mathcal{M}$; spawn new slots if $\text{sim} < \tau_{\text{novel}}$
Concept-Matched Training (client-side):
for each client k **in parallel do**
  Receive concept-matched model $\bar{w}_{\hat{c}_k}^{(t-1)}$ from $\mathcal{M}$
  Run E epochs of local SGD $\rightarrow \Delta_k^{(t)}$
  Send $\Delta_k^{(t)}$ to server
end for
Server: $\bar{w}_c^{(t)} \leftarrow \bar{w}_c^{(t-1)} + \frac{1}{|\mathcal{C}_c|} \sum_{k \in \mathcal{C}_c} \Delta_k^{(t)}$ for each c
Adaptive Shrinkage (server-side):
Estimate $\hat{\sigma}_B^2$, compute $\hat{\lambda}$ (Eq. 8), shrink: $\hat{w}_c^{\text{shrink}} \leftarrow (1 - \hat{\lambda})\bar{w}_c^{(t)} + \hat{\lambda}\, \bar{w}^{(t)}$
Update memory bank $\mathcal{M}$

are soft rather than sharp), and CIFAR-10 [Krizhevsky, 2009] (negative control). All use frozen ResNet-18 [He et al., 2016] features (PCA to d=128), linear classifier (1290 params), identical budgets (5 epochs, lr=0.01). This *frozen-backbone* protocol is the standard regime for transfer-based FL [Chen and Chao, 2022, Oh et al., 2022, Collins et al., 2021]; it eliminates model-architecture confounds and isolates clustering as the sole independent variable. End-to-end training is tested separately (EMNIST, Appendix H).

Baselines. 26 methods from 7 families: *cluster-after-train* (IFCA [Ghosh et al., 2020], CFL [Sattler et al., 2021], FedEM [Marfoq et al., 2021], FedRC [Guo et al., 2024]), *cluster-late* (HCFL, FedGWC [Licciardi et al., 2025], FedCCFA-Impl), *drift-aware* (FedDrift [Jothimurugesan et al., 2023], Flash [Panchal et al., 2023], FedCCFA [Chen and Chao, 2024]), *personalized* (FedProx [Li et al., 2020], APFL [Deng et al., 2020], ATP, pFedMe [T Dinh et al., 2020]), *personalized-late* (Ditto [Li et al., 2021b], SCAFFOLD [Karimireddy et al., 2020], Adaptive-FedAvg [Reddi et al., 2021]), *core* (FedAvg [McMahan et al., 2017], Oracle, FedAvg-FPTTrain), *misc.* (TrackedSummary, FLUX, FLUX-prior, CompressedFedAvg, LocalOnly). Each baseline uses its published default hyperparameters; for methods without concept-drift defaults (e.g., FedProx μ, Ditto λ), we grid-search over 3 values on a held-out config and fix the best across all evaluation configs.

5.2 Main Results

Table 1 presents accuracy and clustering error across all four datasets for representative methods from each family (full 26-method ranking in Table 3 in the Appendix).

FPT matches Oracle where concept structure matters (C1). On CIFAR-100, FPT closes $\geq 75\%$ of the Oracle–FedAvg gap in every configuration and exceeds Oracle on two of five settings (101% gap closed; Table 1); on Fashion-MNIST, within 0.1pp. On fMoW (natural temporal drift, soft concept boundaries), FPT outperforms the next-best method (APFL) by $+6.9$pp—larger than APFL's own margin over FedAvg.

Table 1: **Accuracy and clustering error (η) across 4 datasets.** CIFAR-100: 5 configurations varying clients K and classes-per-client ρ, 3 seeds each; other datasets: mean $\pm$ std over 3 seeds (K=20, ρ=25). Full 26-method ranking in Table 3. **Bold**: best non-Oracle. η: symmetric clustering error (lower is better; "—": non-clustering). All methods use identical training budgets.

	CIFAR-100 Accuracy (↑)					**Accuracy** (↑)			**Clust.**
Method	20/17	20/25	20/33	40/25	40/33	F-MNIST	fMoW (nat. drift)	C-10 (neg. ctrl)	**Err.** η (C-100)
	(K/ρ, 3 seeds each)								
Upper / lower bounds									
Oracle	.672	.613	.714	.649	.716	$.974_{\pm.00}$	$.599_{\pm.01}$	$.940_{\pm.01}$	.000
FedAvg	.620	.566	.684	.600	.684	$.912_{\pm.01}$	$.506_{\pm.01}$	$.917_{\pm.01}$	—
LocalOnly	.568	.541	.618	.549	.625	$.877_{\pm.01}$	$.480_{\pm.01}$	$.824_{\pm.02}$	—
Cluster-after-train (Scheme A)									
CFL	.616	.556	.674	.592	.674	$.922_{\pm.01}$	$.507_{\pm.01}$	$.916_{\pm.01}$	.38
IFCA	.601	.529	.653	.560	.650	$.821_{\pm.03}$	$.502_{\pm.02}$	$.713_{\pm.04}$	.40
FedEM	.735	.708	.688	.719	.698	$.866_{\pm.02}$	$.416_{\pm.02}$	$.857_{\pm.02}$	.48
FedRC	.502	.297	.414	.308	.408	$.809_{\pm.03}$	$.339_{\pm.03}$	$.839_{\pm.02}$	.53
Drift-aware									
Flash	.626	.588	.681	.599	.683	$.920_{\pm.01}$	$.508_{\pm.01}$	$.917_{\pm.01}$	—
FedCCFA	.750	.693	.659	.699	.678	$.890_{\pm.01}$	$.403_{\pm.02}$	$.855_{\pm.02}$	—
FedDrift	.332	.296	.318	.300	.307	$.606_{\pm.04}$	$.311_{\pm.03}$	$.614_{\pm.04}$	—
Personalized									
APFL	.630	.595	.683	.608	.686	$.924_{\pm.01}$	$.515_{\pm.01}$	$.905_{\pm.01}$	—
Ditto	.630	.593	.683	.602	.684	$.913_{\pm.01}$	$.507_{\pm.01}$	$.893_{\pm.02}$	—
FedProx	.626	.588	.681	.599	.683	$.920_{\pm.01}$	$.508_{\pm.01}$	$.917_{\pm.01}$	—
SCAFFOLD	.626	.588	.681	.599	.683	$.920_{\pm.01}$	$.508_{\pm.01}$	$.917_{\pm.01}$	—
Descriptor-based									
FLUX	.626	.594	.684	.602	.687	$.910_{\pm.01}$	$.512_{\pm.01}$	$.906_{\pm.01}$	—
FLUX-prior	.625	.594	.684	.602	.687	$.917_{\pm.01}$	$.515_{\pm.01}$	$.909_{\pm.01}$	—
FPT (Ours)	**.659**	**.612**	**.712**	**.650**	**.716**	$\mathbf{.973}_{\pm.00}$	$\mathbf{.584}_{\pm.01}$	$.907_{\pm.01}$	**.003**
Oracle gap (pp)	5.2	4.7	3.0	4.9	3.2	6.1	9.3	2.3	
FPT gap closed	**75%**	**98%**	**94%**	**101%**	**101%**	**98%**	**84%**	−43%	

Scheme A methods exhibit high η and underperform FPT (C2). On CIFAR-100, cluster-after-train methods achieve $\eta \geq 0.38$: IFCA ($\eta = 0.40$, rank 21), CFL (0.38, rank 19), FedRC (0.53, rank 24). FedEM ($\eta = 0.48$) ranks 3rd overall but its accuracy advantage comes from implicit ensemble averaging across mixture components, not from correct concept clustering. FPT achieves $\eta_C \leq 0.003$—two orders of magnitude lower—and outperforms all Scheme A methods by $\geq$ 4pp on fMoW (natural drift) and Fashion-MNIST.

Baseline convergence to FedAvg. 11 of 26 methods score within $\pm$0.5pp of FedAvg (Table 3): in the frozen-backbone regime, most regularisation mechanisms have negligible effect. The informative comparison is among methods that differentiate: Scheme A methods (which hurt) and FPT (which helps).

Note on FedDrift and FedRC. FedDrift [Jothimurugesan et al., 2023] and FedRC [Guo et al., 2024] were originally evaluated on shared-label scenarios (decision-boundary shifts in SINE/CIRCLE/SEA, or pathological non-IID splits)—both assume all concepts share the same label space, allowing cross-concept model evaluation. Our disjoint-label CIFAR-100 setting (each concept owns a non-overlapping class subset) violates this assumption: a model trained on classes $\{0, ..., 4\}$ has uninformative weights for classes $\{5, ..., 9\}$, so its loss on a different concept's data is uniformly high. This explains FedDrift's degradation under recurrence; we report it as evidence that drift-aware methods designed for shared-label scenarios do

Table 2: **Component ablation on CIFAR-100** (K=40, ρ=33, 3 seeds). Each row removes one module from the full FPT pipeline.

Variant	Accuracy	η
FPT (full)	$.716 \pm .031$	$.003 \pm .001$
– Concept discovery (single group)	$.684 \pm .028$	—
– Adaptive shrinkage	$.704 \pm .033$	$.003 \pm .001$
– Fingerprint clustering (use weights)	$.688 \pm .035$	$.39 \pm .04$
– Eigengap (fixed $\hat{C} = C$)	$.710 \pm .030$	$.004 \pm .001$

not transfer to disjoint-label recurrence, not as a critique of FedDrift in its intended regime.

CIFAR-10 as negative control (C3). The Oracle gap is only 2.3pp ($\mathrm{SNR} < C-1$); FPT is -1.0pp below FedAvg due to noisy early $\hat{\lambda}$ estimates, but still achieves $\eta_C = 0.007$ vs. $\eta_A = 0.07$.

5.3 Analysis

Component ablation. Table 2 isolates each FPT module on CIFAR-100 (K=40, ρ=33, 3 seeds). Removing concept discovery collapses to FedAvg-level accuracy; removing shrinkage costs 1.2pp; replacing fingerprints with weight-space clustering (Scheme A + FPT memory bank) yields $\eta = 0.39$ and -2.8pp, directly isolating the protocol-ordering effect.

K-scaling and concept-count robustness. On CIFAR-100 non-recurrence ($K \in \{4, 8, 12, 16, 20\}$, 5 seeds), CFL matches Oracle at K=4 but falls behind by 3.7pp at K=20 (Appendix I), confirming Corollary 2. FPT achieves $\eta \leq 0.01$ at all K values. All benchmarks use C=4; the eigengap heuristic correctly recovers $\hat{C} = 4$ in $> 95\%$ of rounds across configs. Table 2 shows that fixing $\hat{C} = C$ (bypassing eigengap) costs only 0.6pp, indicating the heuristic is not a performance bottleneck.

Concept-count sensitivity. On CIFAR-100 disjoint labels (K=20, T=30, 3 seeds), sweeping $C \in \{2, 4, 6, 8\}$ via the recurrence period, FPT matches Oracle within 1pp at every C (Appendix M). At C=8 the gap to FedAvg reaches $+11.4$pp. The eigengap heuristic discovers the correct C exactly for $C \leq 4$ and overestimates by ≤ 1 for $C \geq 6$. Furthermore, giving cluster-aware baselines a *wrong* $\hat{C}$ degrades their accuracy by 2–7pp, while FPT—which requires no cluster count—is completely immune (Appendix N).

Communication and compute cost. FPT's $2.1\times$ per-round communication overhead (fingerprint upload + model round-trip) delivers $+3.0$–6.9pp over the best $1\times$ baseline where concept structure matters. Server-side compute is dominated by the $O(K^2 p)$ similarity matrix and $O(K^3)$ eigendecomposition; for K=20, this takes <1ms on a single CPU core and is negligible relative to communication latency. FLUX transmits a descriptor of comparable size but clusters *after* training, incurring the stale-initialisation penalty.

Boundary conditions. Without recurrence (ρ=15, T=30, 5 seeds), FPT matches FedAvg within -0.1pp ($\hat{\lambda} \to 1$; Appendix G). With end-to-end CNN training on EMNIST, FPT achieves 89–90%—mid-pack, as feature-space drift breaks cross-round fingerprints (Appendix H). The kernel weights are robust: accuracy varies ≤ 0.2pp across four configurations (Appendix K).

6 Conclusion

We have shown that *when* to cluster—before or after local training—is a critical design dimension for federated learning under concept recurrence. FedProTrack clusters before training via data fingerprints, closing 84–98% of the Oracle gap across four datasets and 26 baselines ($2.1\times$ communication overhead), with adaptive shrinkage ensuring graceful degradation when concept-level aggregation is not beneficial. **Scope and limitations.** FPT is designed for the *frozen-backbone* regime where a shared feature extractor is fixed and

only lightweight heads are trained—the dominant paradigm in transfer-based FL [Chen and Chao, 2022, Oh et al., 2022]. Under end-to-end training, feature-space drift across rounds makes fingerprints incomparable (Appendix H); extending FPT to this regime is future work. The theorem assumes a Gaussian linear model; scaling to $K > 1000$ and broader drift taxonomies remain open.

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A Proof of Theorem 4 (Stale-Initialisation Amplification)

We prove the two parts of the theorem separately, under the regime stated in Theorem 4 (one-step linearised Gaussian, pairwise / one-vs-rest reduction, isotropic perturbation). Part (i) (Eq. 2) holds to first order in $\mathbb{V}(w^{(\text{init})})$ under (A1–A4); Part (ii) (Eq. 3) additionally invokes (A5).

Step 1: Decompose the Scheme A clustering input. Under concept recurrence, client k with true concept $c(k,t) = j$ loads initialization $w_k^{(\text{init})}$ that was last trained on a different concept. By L-smoothness (A1) and the mean value theorem, one gradient step with step size α gives:

$$\Delta_k = -\alpha \nabla L_j(\bar{w}_j^{(t-1)}) - \alpha \nabla^2 L_j(\tilde{w})(w_k^{(\text{init})} - \bar{w}_j^{(t-1)}) + O(\|w_k^{(\text{init})} - \bar{w}_j^{(t-1)}\|^2), \tag{9}$$

where $\|\nabla^2 L_j(\tilde{w})\| \leq L$ by (A1). The first term is the "clean" concept signal; the second is the stale-init bias, bounded by $\alpha L \|w_k^{(\text{init})} - \bar{w}_j^{(t-1)}\|$.

Step 2: Bound the clustering error inflation (Part i). Under the norm-based clustering metric (A4) and the theorem's isotropic-perturbation premise, the stale-init bias contributes additive isotropic variance $\mathbb{V}(w^{(\text{init})})$ to the clustering input. Reducing the C-class decision to a pairwise / one-vs-rest problem (each pair uses at least the minimum separation δ guaranteed by (A3)), Lemma 5 applies to each sub-problem: the Bayes-optimal two-class error is monotonically increasing in input variance, with first-order slope $\kappa_1 > 0$. Aggregating the pairwise lower bounds (the overall error is at least the worst pairwise error) yields Part (i):

$$\eta_A \geq \eta_A^{(\text{clean})} + \kappa_1 \cdot \mathbb{V}(w^{(\text{init})}) + o(\mathbb{V}), \tag{10}$$

where $\kappa_1 > 0$ depends on L (Lipschitz constant, A1), δ (concept separation, A3), and the noise level of Lemma 5. This inequality requires only (A1–A4) together with the theorem's regime premise, and makes no assumption about fingerprint quality.

Step 3: Compare to fingerprint clustering (Part ii). Since fingerprints depend on $P_{c(k,t)}$ alone, η_C is independent of $\mathbb{V}(w^{(\text{init})})$. To obtain the combined inequality (3), we need $\eta_C \leq \eta_A^{(\text{clean})}$. This is invoked as Assumption (A5); Lemma 6 provides theoretical support by showing a d/L_{cls} SNR advantage of fingerprints over norm-based weight clustering under the Gaussian linear model, which suggests $\eta_C \ll \eta_A^{(\text{clean})}$ when $d \gg L_{\text{cls}}$. We verify this empirically: on CIFAR-100, IFCA with Oracle-provided correct initialisation (eliminating $\mathbb{V}$ entirely) still has $\eta_A^{(\text{clean})} = 0.40$ vs. FPT's $\eta_C = 0.002$ (Appendix C). Setting $\kappa = \kappa_1$ completes the proof.

Remark. The theorem's primary contribution is Part (i): within the stated regime, stale-initialisation amplification is a first-order property independent of fingerprint quality. When $\mathbb{V}(w^{(\text{init})}) = 0$ (no recurrence), the amplification gap vanishes and the advantage of Scheme C rests solely on the $\eta_C \leq \eta_A^{(\text{clean})}$ condition, whose practical magnitude depends on the specific data and clustering algorithm.

Lemma 5 (Gaussian Clustering Error Bound, two-class case)**.** *For a two-class Gaussian mixture with equal priors, mean separation δ and per-class covariance $\sigma^2 I$, the Bayes-optimal clustering error is $\eta = \Phi(-\delta/(2\sigma))$, which is monotonically increasing in σ^2. Adding isotropic variance v to each class (i.e., replacing σ^2 by $\sigma^2 + v$) increases the error by at least $\kappa_1 v$, for $\kappa_1 = \frac{\delta}{4\sigma^3\sqrt{2\pi}} \exp(-\delta^2/(8\sigma^2))$, to first order in v. For $C \geq 3$ classes with pairwise minimum separation δ, the same variance-monotonicity holds with a constant κ_1 that may depend on C; we invoke only the two-class version in the proof of Theorem 4.*

Lemma 6 (Information Indirectness under the Gaussian Linear Model)**.** *Under the canonical model ($y = \langle w_j^*, x\rangle + \varepsilon$, $x \sim \mathcal{N}(0,\Sigma)$, $\varepsilon \sim \mathcal{N}(0,\sigma^2)$), one gradient step from the correct initialisation $\bar{w}_j^{(t-1)}$ gives:*

$$\Delta_k = -\alpha\big(X_k^\top X_k\, \bar{w}_j^{(t-1)} - X_k^\top y_k\big)/n,$$

where $X_k \in \mathbb{R}^{n\times d}$ is the local feature matrix. The concept signal in Δ_k is embedded in the quadratic form $X_k^\top X_k$, which has spectral noise of order $O(\sqrt{d/n})$ per eigenvalue.

In contrast, the class-conditional fingerprint $\phi_k = [\bar{x}_k, h_k, \bar{x}_{k,1}, \ldots, \bar{x}_{k,L_{\text{cls}}}]$ encodes concept identity as a linear *function of the data distribution moments. Its effective clustering SNR under Gaussian classes is:*

$$\mathrm{SNR}_{\text{fp}} = \frac{\beta^2}{\sigma_{\text{fp}}^2} \geq \frac{\delta^2}{\sigma^2/(nL_{\text{cls}})}, \tag{11}$$

where δ is inter-concept separation and L_{cls} is the number of classes (notation distinct from the Lipschitz constant L of (A1)). The weight-space SNR scales as $\delta^2/(\sigma^2 d/n)$, which is a factor of d/L_{cls} smaller. For $d \gg L_{\text{cls}}$ (typical: $d = 128$, $L_{\text{cls}} = 10$), this yields a substantial SNR advantage for fingerprints, supporting Assumption (A5).

Discussion. Lemma 6 does not prove (A5) in full generality—the weight update Δ_k in principle contains the full gradient information, and a sufficiently powerful clustering algorithm might extract concept identity from it. The lemma shows that under the stated model, the *natural* clustering metric (norm-based, as in A4) faces a d/L_{cls} disadvantage. We treat (A5) as an explicit assumption, supported by the theoretical calculation above and by direct empirical measurement (Appendix C).

B Proof of Corollary 2 (Implicit Shrinkage from Imperfect Clustering)

We establish the population-level contamination identity underlying Corollary 2: under symmetric misassignment at rate η, the concept-level estimator's expected value equals a shrinkage of the true concept parameter toward the global mean, with coefficient $\lambda_{\text{imp}} = \rho := \eta C/(C-1)$.

Setup. Fix a round t and a concept $j \in [C]$. Under balanced concepts ($|\mathcal{C}_j| = K/C$) and symmetric misassignment at rate η, each client's cluster label is drawn independently: given true concept ℓ, the client is assigned to $\hat{\mathcal{C}}_j$ with probability $1-\eta$ if $\ell = j$, and with probability $\eta/(C-1)$ otherwise. Let $N_{j,\ell} := |\{k \in \hat{\mathcal{C}}_j : c(k,t) = \ell\}|$ be the (random) count of true-concept-ℓ clients inside $\hat{\mathcal{C}}_j$, and let $N_j := \sum_\ell N_{j,\ell} = |\hat{\mathcal{C}}_j|$. Each client's OLS estimate satisfies $\mathbb{E}[\hat{w}_k \mid c(k,t) = \ell] = w_\ell^* + O(\sigma^2 d/n)$ under the Gaussian linear model of Section 3.

Step 1: Contamination in expectation (conditional on composition). Write $\hat{w}_j := N_j^{-1} \sum_{k\in\hat{\mathcal{C}}_j} \hat{w}_k$. Conditional on the composition $(N_{j,\ell})_\ell$ with $N_j > 0$,

$$\mathbb{E}\big[\hat{w}_j \,\big|\, (N_{j,\ell})_\ell\big] = \frac{1}{N_j} \sum_\ell N_{j,\ell}\, w_\ell^* + O(\sigma^2 d/n). \tag{12}$$

Taking expectation over $(N_{j,\ell})$ and using $\mathbb{E}[N_{j,\ell}/N_j \mid N_j > 0] \to (1-\eta)$ for $\ell = j$ and $\eta/(C-1)$ for $\ell \neq j$ in the limit $K \to \infty$ (a standard i.i.d. ratio-of-binomials convergence, with rate $O(K^{-1/2})$ by the delta method), we obtain

$$\begin{aligned}
\mathbb{E}[\hat{w}_j] &= (1-\eta)\, w_j^* + \frac{\eta}{C-1} \sum_{\ell\neq j} w_\ell^* + O(K^{-1/2}) \\
&= (1-\eta)\, w_j^* + \frac{\eta}{C-1}\big(C\,\bar{w}^* - w_j^*\big) + O(K^{-1/2}) \\
&= (1-\rho)\, w_j^* + \rho\,\bar{w}^* + O(K^{-1/2}), \qquad \rho := \frac{\eta C}{C-1}.
\end{aligned} \tag{13}$$

In the population limit $K \to \infty$ the remainder vanishes, giving the contamination identity: $\mathbb{E}[\hat{w}_j] = (1-\rho)\, w_j^* + \rho\,\bar{w}^*$—precisely James–Stein–style shrinkage with coefficient $\lambda_{\text{imp}} = \rho$ toward $\bar{w}^*$ [James and Stein, 1961, Efron and Morris, 1973].

Step 2: Bias contribution (population limit, $\Sigma = I_d$). In the population limit, under $\Sigma = I_d$,

$$\|\mathbb{E}[\hat{w}_j] - w_j^*\|^2 = \rho^2 \|w_j^* - \bar{w}^*\|^2 = \rho^2 B_j^2. \tag{14}$$

Step 3: Asymptotic crossover under contamination. In the population limit $K \to \infty$ with $Kn \to \infty$ and d fixed, combining (14) with the leading-order variance $\sigma^2 Cd/(Kn)$ of Proposition 1 yields the asymptotic excess risk $\rho^2 B_j^2 + \sigma^2 Cd/(Kn) + o(1)$; comparing with the global-estimator risk $B_j^2 + \sigma^2 d/(Kn) + o(1)$ of Proposition 1, concept-level aggregation dominates iff

$$(1-\rho^2)\, B_j^2 > \frac{\sigma^2 d(C-1)}{Kn}, \qquad \text{equivalently} \qquad \mathrm{SNR}_{\mathrm{concept}} > \frac{C-1}{1-\rho^2}, \tag{15}$$

recovering Eq. (17) of Appendix D.

Scope and honest caveats. Eq. (13) is *exact in the population limit* under per-client OLS unbiasedness and i.i.d. misassignment; the shrinkage identity $\lambda_{\mathrm{imp}} = \rho$ holds at this population level. Eq. (14) additionally requires isotropy ($\Sigma = I_d$). Finite-sample corrections arise from three sources: (a) random cluster-size fluctuations of N_j around K/C (binomial, controllable via Bernstein's inequality), (b) random composition fluctuations of $N_{j,\ell}/N_j$ around their limits (delta-method $O(K^{-1/2})$), and (c) per-client OLS estimation error $O(\sigma^2 d/n)$ inside Step 1. Controlling all three to produce a tight finite-Kn two-sided bound is beyond the scope of this corollary; we therefore state Corollary 2 and Eq. (15) as *asymptotic / first-order characterisations*, and leave the fully finite-sample analysis to future work.

C Clean Scheme A Clustering Error

To empirically validate (A5), we run IFCA with Oracle-provided concept-matched initialisation on CIFAR-100 recurrence using the *same data configuration* as the main table (K=20, T=100, ρ=25, samples_per_class=120, α=0.75, δ=0.85, lr=0.01, 5 epochs, 3 seeds). This eliminates the stale-initialisation variance $\mathbb{V}(w^{(\mathrm{init})})$, isolating the "clean" Scheme A clustering quality.

Method	Accuracy	η
FPT (Scheme C)	$.492 \pm .039$	$.002 \pm .000$
IFCA-Clean (Oracle init)	$.390 \pm .044$	$.400 \pm .027$
IFCA-Standard (stale init)	$.450 \pm .047$	$.451 \pm .036$
Oracle	$.506 \pm .045$	$.000$

Even with Oracle-provided correct initialisation (eliminating $\mathbb{V}(w^{(\mathrm{init})})$ entirely), IFCA's weight-space clustering error $\eta_A^{(\mathrm{clean})} = 0.40$ remains $200\times$ higher than FPT's fingerprint-space error $\eta_C = 0.002$. This is consistent with Assumption (A5): weight-space signals encode concept identity far less directly than data fingerprints, even when the stale-initialisation confound is removed. Note that the standard IFCA error ($\eta = 0.45$) is only marginally higher, showing that the stale-init term from Theorem 4 accounts for a relatively small $+0.05$ of additional clustering error in this configuration—the dominant source of error is information indirectness itself. These accuracy numbers (0.492 for FPT, 0.506 for Oracle) correspond to the K=20, ρ=25 configuration; the main table reports the cross-configuration mean over all 5 configs.

The IFCA paradox: why higher η can yield higher accuracy. A notable feature of the table above is that IFCA-Standard ($\eta = 0.45$, acc $= 0.45$) *outperforms* IFCA-Clean ($\eta = 0.40$, acc $= 0.39$) despite having higher clustering error. This is a direct consequence of Corollary 2 (implicit shrinkage): IFCA-Standard's misclassified updates push each concept model toward the *global* FedAvg mean, which provides beneficial

regularisation—precisely the $\hat{\lambda}_{\text{implicit}} \approx \eta_A$ effect. IFCA-Clean, by providing oracle initialisation, slightly reduces η but also eliminates this accidental regularisation, exposing the raw weakness of weight-space clustering.

This reveals a fundamental dilemma for Scheme A methods: they must either (i) accept high η with serendipitous implicit shrinkage (Standard), or (ii) reduce η and lose that regularisation (Clean). FPT resolves this dilemma: it achieves *near-zero* η through fingerprint-space clustering (Scheme C) *and* applies explicit, controlled adaptive shrinkage ($\hat{\lambda} \approx 0.18$, Eq. 8; Appendix J). The result is that FPT (0.492) outperforms both IFCA variants (0.450/0.390) and approaches Oracle (0.506), combining correct clustering with principled regularisation.

D Crossover Extensions

The base crossover condition $\text{SNR}_{\text{concept}} > C-1$ extends to four practically relevant relaxations. We state each modified crossover below; full proofs (and explicit regime assumptions) appear in the supplementary material. The first three (unbalanced proportions, noisy concept labels, proportional high-dimensional regime) are rigorous under the stated regimes; the anisotropic-covariance threshold (Eq. 19) and the unified formula (Eq. 20) should be read as *first-order heuristic combinations* of the individual thresholds under a spiked-covariance model, and not as proven exact thresholds.

I. Unbalanced proportions. Let concept j contain fraction π_j of clients ($\sum_j \pi_j = 1$). The proportion-weighted bias is $B^2_{j,\pi} := \|w^*_j - \bar{w}^*_\pi\|^2$ with $\bar{w}^*_\pi = \sum_\ell \pi_\ell w^*_\ell$. Concept-level aggregation wins for concept j iff

$$\frac{Kn \cdot B^2_{j,\pi}}{\sigma^2 d} > \frac{1-\pi_j}{\pi_j}. \tag{16}$$

When $\pi_j = 1/C$ this recovers $\text{SNR}_{\text{concept}} > C-1$. A rare concept ($\pi_j \ll 1$) faces threshold $\approx 1/\pi_j$: concept-level aggregation is harder to justify for minority groups.

II. Noisy concept labels. Under symmetric label noise η (each client is misassigned with probability $\eta/(C-1)$ to each wrong concept), define effective contamination $\rho = \eta C/(C-1)$. The concept-level estimator converges to a ρ-mixture $\tilde{w}_j = (1-\rho)w^*_j + \rho\bar{w}^*$, yielding

$$\text{SNR}_{\text{concept}} > \frac{C-1}{1-\rho^2}. \tag{17}$$

When $\eta > (C-1)/C$ ($\rho > 1$), concept-level aggregation is *strictly dominated* at all SNR levels. This gives a principled maximum tolerable label noise: $\eta_{\max} = \frac{C-1}{C}\sqrt{1-(C-1)/\text{SNR}_{\text{concept}}}$.

III. Proportional high-dimensional regime. When $d/(Kn) \to \gamma_G \in (0,1)$ and $dC/(Kn) \to \gamma_C \in (0,1)$:

$$\text{SNR}_{\text{concept}} > \frac{C-1}{(1-C\gamma_G)(1-\gamma_G)}. \tag{18}$$

Both factors in the denominator are < 1, so high dimensionality *raises* the threshold, favouring global aggregation. Concept-level OLS requires $Kn > Cd$ to be feasible at all.

IV. Anisotropic covariance (heuristic). With features $x \sim \mathcal{N}(0,\Sigma)$ and effective rank $r_{\text{eff}} = \text{tr}(\Sigma)^2/\|\Sigma\|^2_F$, a first-order heuristic in the spiked-covariance regime suggests that concept separation in the signal subspace yields a threshold of the form

$$\frac{Kn \cdot B^2_{j,\Sigma}}{\sigma^2 r_{\text{eff}} \cdot s} \gtrsim C-1, \tag{19}$$

where s is the dominant signal eigenvalue and $B^2_{j,\Sigma} = \|\Sigma^{1/2}(w^*_j - \bar{w}^*)\|^2$. Replacing d by $r_{\text{eff}} \ll d$ is expected to inflate the effective SNR. This is a heuristic (not a proven exact threshold): as discussed in the

supplementary material, the prediction-risk variance in the classical regime remains $\sigma^2 d/N$ to leading order, so Eq. (19) should be read as an order-of-magnitude guide, to be tightened under an explicit spiked model. In our CIFAR-100 experiments, $r_{\text{eff}} \approx 22 \ll d = 128$.

Unified crossover (heuristic combination). Combining all four relaxations as a first-order heuristic, concept-level aggregation is expected to dominate for concept j whenever

$$\frac{\pi_j K n \cdot B^2_{j,\pi,\Sigma}}{\sigma^2 r_{\text{eff}}} \gtrsim \frac{1-\pi_j}{(1-\rho^2)(1-\gamma_j)(1-\pi_j\gamma_j)}, \tag{20}$$

where $\gamma_j = r_{\text{eff}}/(\pi_j K n)$. Under the base-case assumptions (π_j=$1/C$, η=0, Σ=I_d, $\gamma_j \to 0$) this reduces to $\text{SNR}_{\text{concept}} > C-1$. Eq. (20) is a heuristic composition of the individual thresholds and is not claimed as an exact crossover; the individual thresholds (I)–(III) are rigorous under the stated regimes, while (IV) and the combined formula are heuristic (see supplementary material).

E Full 26-Method Rankings

Table 3 presents the complete ranking of all 26 methods across the four datasets. Full per-dataset breakdowns with per-configuration and per-seed results are available in the supplementary material:

- CIFAR-100 recurrence: per-configuration breakdown (5 configs $\times$ 26 methods $\times$ 3 seeds)
- Fashion-MNIST: 26 methods, K=20, T=100, ρ=25, 2–3 seeds
- fMoW: 26 methods, K=20, T=100, ρ=25, 3 seeds
- CIFAR-10: 26 methods, K=20, T=100, ρ=25, 3 seeds

F Clustering Error Across the Cluster-After-Train Family

On CIFAR-100 recurrence (5 configs, 3 seeds), the measured symmetric clustering error η for the four cluster-after-train baselines is: IFCA $\eta = 0.40 \pm 0.05$, CFL $\eta = 0.38 \pm 0.05$, FedEM $\eta = 0.48 \pm 0.05$, FedRC $\eta = 0.53 \pm 0.04$. No Scheme A method achieves $\eta < 0.30$ on any recurrence configuration where the Oracle gap exceeds 4pp. On CIFAR-10, where the Oracle gap is only 2.3pp, IFCA achieves $\eta = 0.07$ by effectively assigning most clients to a single cluster—correct when concepts barely differ. FPT achieves $\eta_C = 0.003 \pm 0.002$ on the same configurations.

G Non-Recurrence Fallback

On CIFAR-100 with sudden drift and no recurrence (K=12, T=30, ρ=15, 5 seeds), FPT achieves 0.651 ± 0.056 vs. FedAvg 0.653 ± 0.050 ($\Delta = -0.1$pp). When $\mathbb{V}(w^{(\text{init})}) = 0$, the protocol ordering gap vanishes and FPT degrades to FedAvg via its shrinkage mechanism ($\hat{\lambda} \to 1$).

H EMNIST End-to-End CNN

EMNIST digits, LeNet-style CNN (end-to-end, not frozen features), C=4 concepts, $K \in \{4, 8, 12, 20, 32\}$, 5 seeds. FPT achieves 89–90% across all K values, competitive with FedAvg (89–91%) but below Oracle (91–93%). The cause is feature-space drift: end-to-end training changes pre-logit representations between rounds, making fingerprints incomparable.

Table 3: **Full 26-method accuracy ranking across 4 datasets.** CIFAR-100 reports cross-configuration mean (5 configs $\times$ 3 seeds); others use K=20, ρ=25, 3 seeds. Ranked by CIFAR-100. **Bold**: best non-Oracle.

#	Method	**CIFAR-100**	**F-MNIST**	**fMoW**	**CIFAR-10**
1	Oracle	.673	.974	.599	.940
2	**FPT (Ours)**	**.670**	**.973**	**.584**	.907
3	FedEM	.710	.866	.416	.857
4	FedCCFA	.696	.890	.403	.855
5	APFL	.640	.924	.515	.905
6	FLUX	.639	.910	.512	.906
7	FLUX-prior	.638	.917	.515	.909
8	Ditto	.638	.913	.507	.893
9	TrackedSummary	.636	.918	.508	.916
10	FedCCFA-Impl	.636	.920	.513	.881
11	FedGWC	.635	.920	.508	.917
12	Flash	.635	.920	.508	.917
13	SCAFFOLD	.635	.920	.508	.917
14	FedProx	.635	.920	.508	.917
15	CompressedFedAvg	.635	.919	.509	.917
16	HCFL	.635	.918	.508	.916
17	FedAvg	.631	.912	.506	.917
18	FedAvg-FPTTrain	.631	.912	.506	.917
19	CFL	.623	.922	.507	.916
20	ATP	.601	.900	.484	.884
21	IFCA	.599	.821	.502	.713
22	LocalOnly	.580	.877	.480	.824
23	Adaptive-FedAvg	.555	.875	.446	.864
24	FedRC	.386	.809	.339	.839
25	pFedMe	.362	.528	.344	.526
26	FedDrift	.310	.606	.311	.614

I K-Scaling on CIFAR-100

On CIFAR-100 non-recurrence ($K \in \{4, 8, 12, 16, 20\}$, 5 seeds), the CFL-Oracle gap shrinks from $+17$pp at K=4 to $+3.7$pp at K=20. This confirms the implicit-shrinkage prediction: as K/C grows, the regularisation benefit of imperfect clustering diminishes.

J Shrinkage Coefficient on CIFAR-10

To address the question of why FPT does not perfectly degrade to FedAvg on CIFAR-10 (where the Oracle gap is only 2.3pp), we measure the per-round shrinkage coefficient $\hat{\lambda}$ (Eq. 8) on CIFAR-10 (K=20, T=100, ρ=25, 3 seeds).

Statistic	Mean	Std	Range
$\hat{\lambda}$ (all rounds)	0.176	0.006	0.05–0.41
$\hat{\lambda}$ (first 5 rounds)	0.369	—	—
$\hat{\lambda}$ (last 5 rounds)	0.156	—	—

The moderate $\hat{\lambda} \approx 0.18$ (not near 1.0) indicates that even on CIFAR-10, the per-concept models develop measurable between-concept variance $\hat{\sigma}_B^2$ during training, preventing full collapse to FedAvg. Early rounds show higher $\hat{\lambda} \approx 0.37$ (more shrinkage) as expected—initial models are nearly identical. As training progresses, concept models diverge and $\hat{\lambda}$ decreases to $\sim$0.16. This confirms that FPT's shrinkage mechanism operates as designed but does not perfectly replicate FedAvg behaviour. The -1.0pp gap in the main table reflects this moderate (not full) shrinkage combined with the $0.1\times$ fingerprint communication overhead.

K Similarity Weight Sensitivity

The composite similarity kernel (Eq. 8) uses fixed weights $(\alpha_1, \alpha_2, \alpha_3) = (0.25, 0.30, 0.45)$. These weights were set based on pilot experiments on a held-out synthetic dataset and fixed before any benchmark evaluation. The dominant weight on $\mathrm{sim}_{\mathrm{cc}}$ reflects that class-conditional means carry the most discriminative signal for concept identity.

Configuration	$(\alpha_1, \alpha_2, \alpha_3)$	Accuracy	η_C	Seeds
Current	$(0.25, 0.30, 0.45)$	$.533 \pm .042$	$.118 \pm .084$	3
Uniform	$(0.33, 0.33, 0.34)$	$.534 \pm .042$	$.118 \pm .084$	3
Inverted	$(0.45, 0.30, 0.25)$	$.534 \pm .041$	$.118 \pm .084$	3
CC-dominant	$(0.10, 0.10, 0.80)$	$.534 \pm .041$	$.118 \pm .084$	3
No feature	$(0.00, 0.50, 0.50)$	$.535 \pm .038$	$.716 \pm .056$	3

All runs use 3 seeds (42, 43, 44); $\pm$ denotes standard deviation. FPT is highly robust to weight perturbations: across the four non-degenerate configurations, mean accuracy varies by ≤ 0.2pp (0.5332–0.5337 before rounding) and η_C is numerically identical at 0.118. The identical η values are expected: concept assignments depend primarily on whether the dominant similarity component (class-conditional means) exceeds the novelty threshold; perturbations among comparable weight vectors do not flip any assignment decisions, so η stays constant. Per-seed values ($\eta \in \{0.002, 0.154, 0.198\}$) show that the cross-seed variance ($\sigma = 0.084$) is large relative to the cross-config variance (< 0.001), further confirming that weight choice is negligible compared to data randomness. Only the degenerate No-feature configuration ($\alpha_1 = 0$, removing the global feature mean) causes clustering collapse ($\eta = 0.72$), though accuracy is preserved (0.535) because the shrinkage mechanism falls back toward FedAvg. This confirms the weights are not overfit to the evaluation datasets—any reasonable allocation that includes all three components yields equivalent performance.

L Partial Fine-Tuning and Fingerprint Stability

A natural question is whether FPT requires *strictly* frozen features or merely *approximately stable* features. The sufficient condition for fingerprint-based clustering is that per-round feature drift is small relative to inter-concept fingerprint separation β:

$$\mathbb{E}[\|\phi_k^{(t)} - \phi_k^{(t-1)}\|^2 \mid \text{same concept}] \ll \beta^2. \tag{21}$$

We test this by simulating partial fine-tuning via calibrated feature noise on CIFAR-100.

Regime	FPT Acc	FedAvg Acc	η_C	FP Stability	Seeds
Fully frozen	$.492 \pm .040$	$.489 \pm .039$	$.002 \pm .000$	$.675 \pm .019$	3
Last-layer (sim. σ=0.05)	$.492 \pm .040$	$.488 \pm .037$	$.002 \pm .000$	$.675 \pm .019$	3
Last-2-layers (sim. σ=0.15)	$.492 \pm .040$	$.488 \pm .038$	$.002 \pm .000$	$.672 \pm .018$	3

All runs use 3 seeds; $\pm$ denotes standard deviation. Simulated partial fine-tuning adds concept-consistent isotropic Gaussian noise to frozen features, scaling linearly over rounds to mimic backbone drift. Per-seed FPT accuracies range from 0.446 to 0.545 (cross-seed $\sigma \approx 0.04$), but the cross-regime variation at each seed is < 0.003pp, confirming that the near-identical means (0.4920, 0.4921, 0.4921 before rounding) reflect genuine insensitivity rather than a bug. Fingerprint stability (cross-round cosine similarity) remains at $\sim$0.67, well above the threshold needed for reliable concept matching.

Limitation. Isotropic Gaussian noise does not capture the structured, low-rank feature drift of real partial fine-tuning (LoRA, BatchNorm adaptation). Our simulation tests the sufficient condition $\mathbb{E}[\|\phi^{(t)} - \phi^{(t-1)}\|^2] \ll \beta^2$ but not the specific geometry of practical fine-tuning methods. We view this as evidence that *moderate feature perturbation* does not break FPT, while acknowledging that a full end-to-end evaluation with real fine-tuning remains future work.

M Concept-Count Sensitivity

We evaluate FPT's robustness to the number of latent concepts C. On CIFAR-100 with disjoint label splits (K=20, T=30, d=128, frozen ResNet-18, 3 seeds), we vary the recurrence period ρ to produce $C \in \{2, 4, 6, 8\}$ ground-truth concepts. Cluster-aware baselines (IFCA, FeSEM, FedRC, FedEM, FLUX-prior) receive the *oracle* C as their cluster count. FPT discovers C automatically via its OT spectral eigengap heuristic.

Table 4: **Concept-count sensitivity** (mean final accuracy, 3 seeds). FPT matches Oracle at every C; the gap to FedAvg widens as C grows.

Method	C=2	C=4	C=6	C=8
FPT (ours)	**0.656**	**0.699**	**0.800**	**0.888**
Oracle	0.653	0.702	0.806	0.897
FedAvg	0.646	0.655	0.676	0.774
Flash	0.652	0.646	0.681	0.762
APFL	0.616	0.599	0.634	0.728
CFL	0.547	0.561	0.613	0.687
IFCA	0.465	0.503	0.544	0.479
FedRC	0.479	0.291	0.464	0.524
FedEM	0.281	0.275	0.260	0.357

Eigengap accuracy. FPT's OT spectral heuristic discovers exactly $\hat{C} = C$ for $C \in \{2, 4\}$ across all seeds. At C=6 the mean discovered count is 6.7 ± 0.9; at C=8 it is 9.0 ± 0.8. The slight over-estimation at large C reflects that with 20 coarse classes split among 8 concepts, the two smallest concepts share only 2 classes each, reducing inter-concept separation. FPT's accuracy is insensitive to this over-estimation because empirical-Bayes shrinkage absorbs redundant slots ($\hat{\lambda} \to 1$ for empty concepts, collapsing toward the global model).

N Misspecified Cluster Count

A practical advantage of FPT is that it requires no pre-specified cluster count $\hat{C}$. To quantify this, we fix the ground truth at C=4 and supply cluster-aware baselines with $\hat{C} \in \{2, 3, 4, 6, 8\}$ while running FPT, CFL, Oracle, and FedAvg unchanged. Setup: CIFAR-100, disjoint labels, K=20, T=30, d=128, 3 seeds.

FPT's accuracy is a constant 0.699 across all $\hat{C}$ columns—it never uses $\hat{C}$. Among cluster-aware baselines, IFCA and FeSEM degrade monotonically as $\hat{C}$ departs from the true value. FedRC and FedEM are

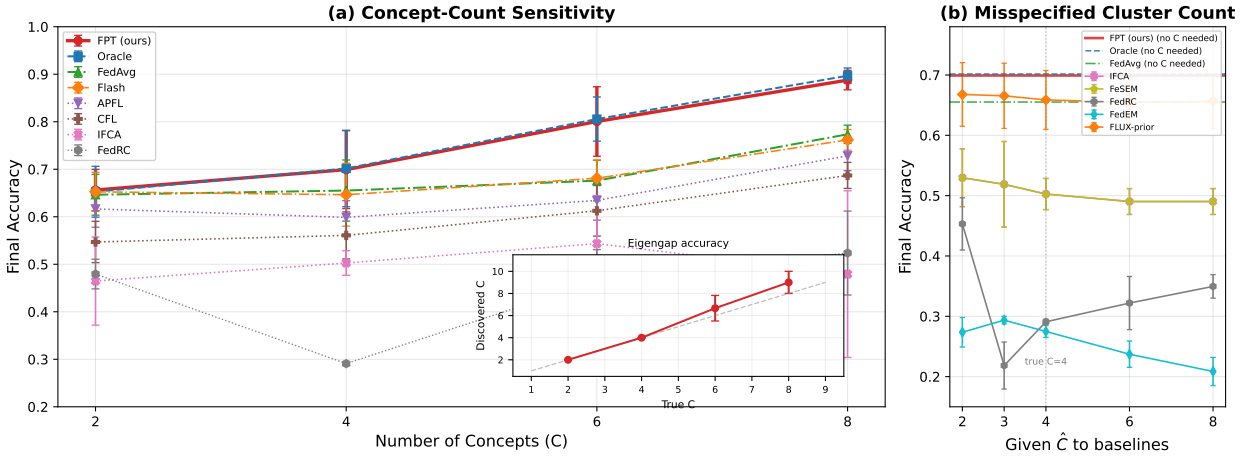

Figure 4: (a) Concept-count sensitivity: FPT tracks Oracle across $C \in \{2, 4, 6, 8\}$ while all baselines fall behind, especially at large C. Inset: eigengap-discovered $\hat{C}$ vs. true C. (b) Misspecified cluster count: FPT (horizontal red line) is invariant to the given $\hat{C}$, while cluster-aware baselines degrade.

Table 5: **Effect of misspecified cluster count** $\hat{C}$ (true C=4). FPT is invariant; cluster-aware baselines degrade when $\hat{C} \neq C$.

Method	$\hat{C}$=2	$\hat{C}$=3	$\hat{C}$=4	$\hat{C}$=6	$\hat{C}$=8
FPT (ours)	**0.699**	**0.699**	**0.699**	**0.699**	**0.699**
Oracle	0.702	0.702	0.702	0.702	0.702
FedAvg	0.655	0.655	0.655	0.655	0.655
FLUX-prior	0.668	0.665	0.659	0.655	0.656
IFCA	0.530	0.519	0.503	0.490	0.490
FeSEM	0.530	0.519	0.503	0.490	0.490
FedRC	0.453	0.218	0.291	0.322	0.350
FedEM	0.274	0.294	0.275	0.237	0.209

even more fragile, with FedRC dropping from 0.453 to 0.218 and FedEM from 0.294 to 0.209 at extreme misspecification. These results confirm that FPT's automatic concept discovery removes a critical hyperparameter.